

STRUCTURING NUMBERS

The Structuring Numbers Instructional Sequence (SN) is designed for first grade to follow the Patterning and Partitioning (P&P) sequence. For organizational purposes we have divided the Structuring Numbers sequence into three parts. A separate video group describes the instructional activities in each of these parts. A fourth video group describes the teacher's role in implementing the sequence successfully.

Notes to Accompany Video Group 1: Overview and Structuring Numbers Part I

In this first video group, we provide an overview of the sequence and then describe Part I of the sequence in detail. We devote a separate video group to each of Parts II and III of the sequence.

Overview

This sequence is based on using the "inquiry approach" to mathematics instruction that has been described in the Patterning and Partitioning videos. We remind teachers that the inquiry approach we advocate is based on establishing the following classroom norms: Students are expected to:

- Solve problems in personally meaningful ways.
- Explain their thinking and reasoning to others.
- Listen to and attempt to make sense of other's thinking and reasoning.
- Ask questions and raise challenges if they do not understand or if they disagree.
- Persist on personally challenging problems.

In addition an important mathematical norm is that:

- Students are expected to give explanations in terms of what is mathematically real to them.

In other words, explanations that are only about rules and procedures are not acceptable.

The SN sequence follows the P&P instructional sequence. In most 1st grade classrooms the SN sequence forms the basis for math instruction for the second nine weeks of the school year. While it may appear that the instructional goals of this sequence can be accomplished in less time, our experience after using it in many classrooms is that the advances in conceptual development that take place as this sequence progresses are foundational for all later work with arithmetic. For this reason, we urge teachers to take the time necessary for children to build these essential ways of numerical reasoning. They form the foundation for fluency with numbers to 20.

This instructional sequence can also be used productively in a curtailed fashion in the first four to six weeks of the second grade year. Also, some kindergarten teachers that use the P&P sequence with their students introduce Part I of the SN sequence at the end of the kindergarten year.

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The SN sequence focuses on numbers between 10 and 20. Specifically, it is designed to support children's development of:

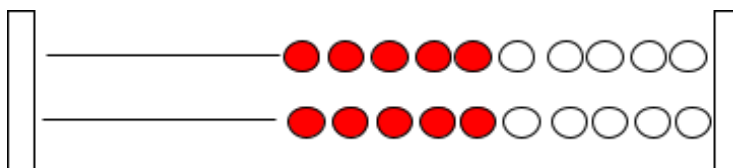
- Grouping strategies for numbers to 20.
- Thinking strategies
- Increasingly sophisticated quantitative interpretations of additive situations.

A child uses a grouping strategy if he splits the numbers into groups that he recombines in efficient ways. For example, the child might reason that $8 + 7$ is 15 because, "I can take 2 out of the 7 and add it to 8 to get 10, then add on the remaining 5 to get 15.

As we explain in the video group on the teacher's role, the sequence also contributes to children's development of thinking strategies. A child uses a thinking strategy when he uses something he just knows or has just figured out to solve the task at hand. For example, the child might say, "I know that $6 + 6$ is 12 so $6 + 7$ has to be just one more. It has to be 13."

In addition, this sequence supports children's development of increasingly sophisticated quantitative interpretations of additive situations. By additive situations we mean addition tasks, subtraction tasks, missing addend tasks, and missing subtrahend tasks.

The sequence is based on the scenario of a double-decker bus and makes use of a device called the Arithmetic Rack (AR) that looks like this.



The arithmetic rack comes from work at the Freudenthal Institute in The Netherlands, where it is called by the Dutch name Rekenrek.

In the same way that the P&P instructional sequence used the single ten-frame as a visual tool to organize and promote mental imagery for numbers to 10 based on doubles and fives, here the AR is used as a way to organize quantities to 20. The structure of the rack promotes imagery for numbers to 20 around doubles, fives and ten. The beads on the rack are movable. They can be moved from right to left and vice versa.

Initially students use their racks to solve tasks. As the sequence progresses, the rack serves as a support for mental imagery and development of number relationships that children can use as a means for reasoning even in the absence of visual support material. We will give an overview of how the sequence progresses before we get into the details of the instructional activities

Based on our experience with this sequence in a large number of classrooms we are able to outline a possible path that the learning of the class as a whole might take. By that we mean

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the types of activities and ways of reasoning that characterize the class as a whole. Such a path is called a learning trajectory.

Learning trajectory for the Structuring Numbers Instructional Sequence:

- Make configurations on the AR.
- Describe configurations on the AR using 5's, 10's, and doubles.
- Reason in terms of groups to solve additive tasks while using the AR.
- Reason in terms of numerical relations to make and evaluate configurations.
- Reason purely numerically to solve additive tasks.

Specifically, in this case, initially we see the class *making* configurations on the rack. The activity then shifts to *describing* configurations in terms of 5's, 10's, and doubles. Another shift occurs when students' activity is characterized by *reasoning in terms of groups* to solve additive task using the rack; Next to *reasoning in terms of numerical relations* to make and evaluate configurations, and finally to *reasoning purely numerically* to solve additive tasks. That is, by the end of the sequence, the rack has dropped out and reasoning is purely numerical.

We want to emphasize that the learning trajectory is not a description of how each individual child will progress, but rather how the class, as a whole, progresses. There is variability within each class. But seen, as a whole, the trajectory lays out a general idea of how the learning progresses.

Stated in less theoretical terms children:

- Learn efficient ways to add and subtract numbers to 20.
- Learn how teen numbers relate to 10,
e.g. $10 + 3 = 13$ $16 - 10 = 6$
- Develop mental imagery for numbers to 20 so concrete materials are no longer needed.

I want to explain in more detail what I mean by each of these.

- Children develop efficient ways to add and subtract numbers to 20.
What do we mean by efficient? In our instructional sequences, "efficient" always means using non-counting approaches. Counting is considered inefficient. Using non-counting methods is considered efficient. Further, non-counting approaches, such as thinking and grouping strategies, are applicable to larger numbers as well. In this sense they are ways of REASONING. So, rather than simply learning mathematical facts, children are learning to reason mathematically.
- Children learn how teen numbers relate to 10.
The number 10 is the basis of our entire number system. It plays a special role. However, when children first learn about number there is no reason for them to treat 10 as special—except perhaps that they have 10 fingers on their hands. So part of what has to happen in children's early development of number is that they have to learn that 10 is "special." Their first encounters with that come with the

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teen numbers and learning the relationship between the teen numbers and the numbers under 10. (The n-teen theorem.)

- Children develop mental imagery for numbers to 20 so concrete materials are no longer needed.

In the same way that a goal of the P&P instructional sequence is to foster children's development of patterns and mental imagery for numbers to 10, one of the goals of the SN sequence is that students develop mental imagery for numbers to 20—not just any imagery, but imagery that is centered around 5's, 10's, and doubles. Once they have mental imagery, children can reason independently of concrete materials.

This instructional sequence is comprised of three parts.

I. Becoming efficient at reading the rack

This part of the sequence is about looking at the rack that shows a certain number of beads and having an efficient way to figure out the quantity shown without counting all of the beads by ones. The emphasis is on using the colors of the beads and the structure of the rack to figure out the quantity shown.

II. Becoming efficient with adding and subtracting using grouping strategies prompted by the rack

As with the first part of this sequence, here the rack is an integral part of the tasks. First children use it as they solve tasks. Then they no longer use the rack themselves but the teacher provides an image of the rack to support children's reasoning.

III. Becoming efficient at additive tasks without the rack, i.e., becoming independent of concrete material and of visual support material

Here children solve problems in a variety of contexts and problems posed in purely numerical settings without using the rack and without having visual images of it to support their reasoning.

Each of these three parts involves several instructional activities. The details of Part I are included in the first video group and described in this set of notes. The details of Parts II and III comprise the next two video groups and are each described in a separate set of accompanying notes.

Part I: Becoming Efficient With READING the Rack

In this part of the sequence, children become familiar with the arithmetic rack as a device that can be used to show quantities up to 20. They learn the conventions of using the rack and become efficient at using the color configurations of the beads, rather than counting, as a basis for reasoning about quantities.

While Part I of the sequence is relatively short in comparison to part II, it is comprised of a number of activities that form a gradual shift in complexity. We list them here:

1. Introductory activities

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2. Show this many people on the bus - shift to focus on efficiency
3. Figure out how many people are on the bus when the configuration is specified
 - a. Partner version of activity 3
4. Rack flash
 - a. Partner version of rack flash
5. Rack bingo
6. Rack flash variation
 - a. Partner version of rack flash variation

In the first several of these activities children use their racks to complete the tasks. Then there is a shift and students no longer use their racks but the teacher shows (or flashes) images of the rack. These visual images are designed to promote children's development of mental imagery for numbers to 20 that are doubles-referenced, five-referenced, and ten-referenced.

You will see this progression and shift in use of the rack as we describe the activities in detail. After we describe the activities, we will provide a suggested timeline for how to progress through this part of the sequence and also make suggestions for how to combine activities to form productive lessons.

1. Introductory Activities

There are three introductory activities:

- Children explore the rack.
- Introduce the double-decker bus scenario.
- Introduce the conventions of the rack.

This first instructional activity is an introductory one and is used only once. The purpose is to introduce the sequence to children. That means introducing the arithmetic rack, the underlying scenario, and the conventions of using the rack.

Children explore the rack. The first step is for children to explore the rack. What do we mean by exploring the rack? As with any tool or device, it is useful to begin by giving students an opportunity to touch it, look it over, move it around, and then talk about what they notice. So each child is given a rack and a few minutes of exploration time. The teacher then initiates a discussion about what children noticed. Typically they say things about the materials used, for example, that the beads are made of wood (or plastic), that the beads are on two rods, that the beads move, that there are two colors of beads with some of each color on both rods, (in our discussion here we talk about red beads and white beads), and so on. It is important that the discussion includes talking about how many beads there are of each color; how many on each rod; how many in all.

Scenario. As part of the introduction to these activities, it is also necessary to introduce the underlying scenario. The scenario is that of a double-decker bus. The

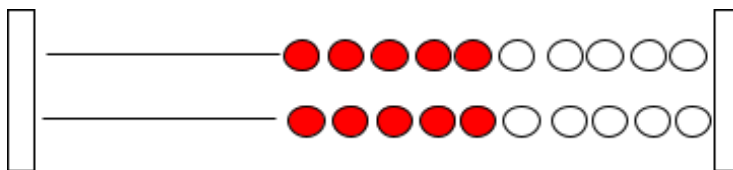
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scenario provides a basis for posing tasks and for children's interpretation of the tasks. For this reason it is important that the teacher takes sufficient time to make the scenario real to the students. They should be able to imagine people riding on the bus, more people getting on the bus or people getting off the bus.

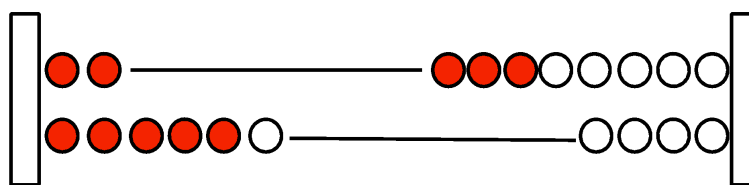
In our experience, teachers typically show great creativity in how they introduce the scenario. Many show a picture of a double-decker bus and talk with the class about who has ridden on a bus like that. The important point is that the scenario becomes realistic to students and they can imagine "acting" within the scenario. They can imagine trying to figure out how many people are on the upper deck and how many are on the lower deck of the bus. Their imaginary actions will form the basis for how they use the rack for reasoning.

Conventions. The teacher ensures that students all place their racks on their desks in the same way so that discussions about beads and colors of beads have uniform meaning. Beads start off at the right and the rack is placed so the white beads are to the right of the red beads.



Starting position of the rack.

Once the scenario is introduced the teacher explains that the arithmetic rack will be used to keep track of *the number of people* on the upper and the lower decks of the bus. This is done by moving that number of beads over to the left on the two rods. For example, to show that there are two people on the upper deck and 6 people on the lower deck we move the beads as shown here.



Shows that the number of people on the upper deck is 2
and the number of people on the lower deck is 6.

It is important to emphasize that the beads show the *number of people* on the bus, not the people themselves. This is an important theoretical point. The rack is not a pretend bus and

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the beads are not pretend people. Rather the beads show the *number of people* on each deck of the bus.

Why does this matter? Well, what if the beads represented people. Suppose the leftmost bead on the upper rod represented a person, say, Sam, and he wanted to get off the bus. Because of the way the rack is made, we are not able to move that bead back over to the right because there is another bead in the way that would have to be moved first. In other words, Sam would not be able to get off the bus. So there would be a problem with the scenario.

Another way to say it is this. The rack is not a pretend bus but is a representation that shows quantity, not people. In this way by using the rack we are already moving above the scenario to focus on number, on quantity, not individual people on a bus.

Having said that, in my experience if the teacher is not careful to make this distinction between number of people and actual people, but conflates the two, the instructional sequence nevertheless is highly productive. I have never seen a child perplexed about “Sam getting off the bus.”

2. Show This Many People on the Bus – Shift to Focus on Efficiency

After the initial introductory activities of exploring the rack, introducing the scenario, and establishing the conventions of the rack, the next instructional activity is to have children use their arithmetic racks to show some number of people on the bus, doing so in any way they like. This activity might be used on the first day immediately after the introductory activity. Or it might be used on the second day, depending on how much time is devoted to the introduction.

Example. Show that there are 8 people on the bus.

The children’s task is to move 8 beads to the left side of the rack in any way they choose. There is no specification of how many people are on the upper deck and how many are on the lower deck.

Before we talk about what children do, I want to explain the purpose (or purposes) of this task. One purpose is to ensure that children understand how to use the rack and how it links to the scenario. In other words, we want to ensure that children understand the task as one of moving 8 beads from the right to the left.

Another purpose is to begin to focus the discussion on the mathematical concepts of interest. We’ll explain what we mean by that when we look at what children do.

What do children do?

- Some children move beads to the left one at a time while counting by ones.
- Some move groups of beads taking advantage of number combinations they know.

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- Some move groups of beads taking advantage of the color groupings of the beads.

The teacher is able to see the variety of ways the children move the beads to accomplish the task of showing that there are 8 people on the bus.

Some children will move beads one at a time while counting by ones. Often these are our least sophisticated children, perhaps some that have not yet developed concepts of composite units.

Some children will move beads in groups taking advantage of number combinations they already know. For example, by this time in the school year many children will just know that 4 and 4 are 8. And they might move groups of 4 beads at once. For example, they might move a group of 4 beads to the left on the top rod and do the same on the bottom rod.

Some children will move all of the red beads on one of the rods and then move 3 more beads, either on the same rod or on the other rod. These children might just know by now that 5 and 3 more make 8.

Another way to say it is that what children do reflects their current conceptual understandings, in particular those understandings they developed through the P&P sequence. Even so, since this is the first task in which children use the AR, some children move beads one at a time even if they already have some number combinations well in hand.

After the children have moved their beads, the teacher engages the class in a discussion. The question is not, "HOW did you move the beads," but is "How did you show that 8 people are on the bus? How many did you show on the top and how many on the bottom?" In other words, it's not what the children *did* but what they *have*.

The purpose of this question is to generate reflection. (Compare to finger patterns where children were asked how many on each hand?) To answer the question, the child has to look back at the result of his actions and describe the results. The child that moved the beads while counting by ones might now have to go back and count the beads on each rod to figure out how many he showed on each deck of the bus. However children that moved groups of beads typically use that information as they answer the question.

Now, here is an important point. When children say that they moved groups of beads, say someone said that he moved 4 beads on each rod, "I moved 4 on the top and 4 on the bottom," the teacher can then probe to ask, "How did you know that was 4? Did you have a way to know without counting the beads by ones?" The child might reply that he moved all but one of the red beads on each rod because there are 5 red beads in all so he needed 1 fewer. Similarly, a child that moved all of the red beads on one rod and three more might say that he knew that moving all the red beads shows 5, etc. What is happening here is that the discussion is moving toward using the color groupings of the beads to describe efficient ways to figure out the quantities. So that brings us to the next point.

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And that point is that one purpose of this task is to *initiate the shift in the focus of the discussion*. Initially the discussion focuses on which beads the children used, that is how many are on the top and how many are on the bottom, but it quickly *shifts to a focus on efficiency*, that is on descriptions based on groups of beads. That shift typically happens within the very first task posed. A major purpose of this task is to get to that shift. For this reason, this task is used only a very few times. The teacher might pose several tasks of this type in one lesson and then in the next lesson move on to the next instructional activity in the sequence.

Having said that, the teacher has to remain fully cognizant of the fact that, even as the instructional sequences progresses, our least sophisticated children might (will) continue to move and count beads by ones for some time. As they participate in the discussions and listen to what others say, they too, over time, will advance to using groups of beads. Remarks by the teacher, such as, “Do you have a way to figure that out without counting?” or “Can you use the colors of the beads to help you figure that out?” are useful in that regard.

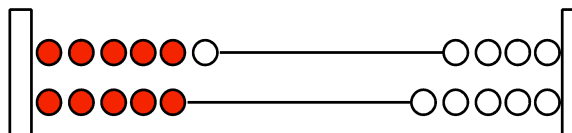
3. Figure Out How Many People Are on the Bus When the Configuration is Specified

The teacher specifies the configuration and the children’s task is to:

- Create the configuration on their own racks.
- And figure out how many people are on the bus.

This third task in the sequence is one that the teacher will use a number of times, over several days. In this task the teacher specifies the configuration of the beads. The children’s task is to show what the teacher specified and then to figure out how many people are on the bus, as shown by that configuration. Children are asked to try to do it without counting. “See if you can use the colors of the beads to help you figure out how many people are on the bus.”

For example, the teacher might project a configuration on the screen



and say, “Make yours look like mine.” (Note: This is not a flashing task. The image is shown and left visible.)

Before we discuss this example in detail, notice how this instructional activity is different from the previous one. In the previous activity children were told the number of people on the bus and their task was to show that using their racks in any way they chose. Here,

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children begin by moving the beads on their racks to show the configuration the teacher has and they are then asked to figure out the number of people on the bus.

The initial part of this task is interesting in and of itself. To make his rack look like the teacher's, the child has to essentially "copy" what the teacher has shown. How might he do that? What might he be thinking? How might that influence how he figures out how many people are on the bus?

In this case, for example, a child might notice that all of the red beads have been moved to the left and also one white bead on the top. The child might move his beads in exactly that way, that is, move all the red beads, and move one white bead on the top. A child can do so without having any idea of how many beads he moved. To figure out how many people are shown on the bus, this child might then count the beads by ones to get 11.

Another child might move the beads in the same way but might be thinking something entirely different. He or she might be thinking, "I have to move all 5 red beads on the top and all 5 red beads on the bottom and also one white bead on the top." This child might figure out that 11 people are shown on the bus by reasoning, "I know 5 and 5 make 10 and 1 more is 11. There are 11 people on the bus."

Yet another child might move the 5 red beads on the top, one white bead on the top, and then the 5 red beads on the bottom. This child might then figure out how many people are on the bus by saying, "There are 6 on the top" and then count on, "7, 8, 9, 10, 11" while pointing to the beads on the bottom, one by one.

The class discussion that accompanies this task focuses on how the children figured out how many people are on the bus and not on the order in which the child moved the beads. Of course, some children might talk about how they moved the beads as part of their explanation of how they figured out how many people are on the bus. For example, the child that thought about the red beads as 5 might say, "I moved 5 red beads on the top and 5 on the bottom, that's 10 and one white one is 11."

When children explain that they figured out how many are on the bus by counting (by ones), it is useful for the teacher to ask, "Does anyone have a way to figure it out without counting?" Or, "Can you use the colors of the beads to help you figure it out without counting?" In this way the teacher acknowledges that she is aware that some of the children still need to count but that figuring it out without counting is more efficient and is a goal of this instructional sequence. After all, the teacher does have a mathematical agenda and it is her responsibility to advance that agenda. One way to do so is to acknowledge when a child counted but not devote time to having the child repeat that counting activity for the class.

Which tasks to pose? We've already said that one goal of this part of the instructional sequence is that children can figure out how many people are on the bus by capitalizing on the color configuration of the beads rather than counting by ones. That

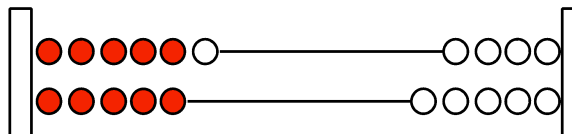
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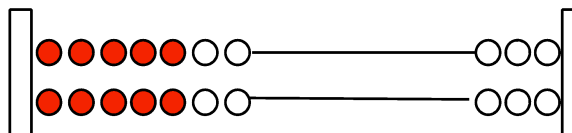
means that teachers have to be deliberate about which configurations they show. Not all configurations lend themselves to using the colors to help figure out the quantity.

Examples of productive tasks. Consider these examples:

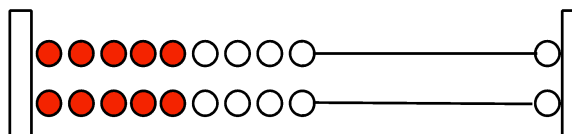
Example 1:



Example 2:



Example 3:



We have already discussed example 1. In our discussion of that example, we have already indicated that the way children figure out how many people are on the bus is often influenced by how they move the beads to show the configuration on their own racks.

In example 2, children that move the red beads on each rod first, and then move the white beads on each rod, might already have a concept of 5 as a composite unit. Such children might reason that 5 and 5 make 10 and then count on, “11, 12, 13, 14” as they point to the white beads. Children that reason this way are often totally unaware that there are 7 on each deck of the bus.

I want to make a comment here about the fact that I said that such children might count on by ones from 10 to get to 14. This might surprise teachers that are new to using instruction of this type. Wouldn't children just reason that 10 and 4 more are 14? The answer is, “No, most first graders would not do that.” In fact, most children at this point in grade 1 do not “just know” that 10 and a small number gives the relevant teen number, that is, 10 and 3 are 13, 10 and 7 are 17, etc. Coming to “just know” this relationship between 10 and a small number and the corresponding teen number is one of the conceptual advances the majority of children make as this sequence progresses.

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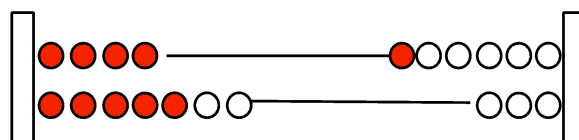
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In both examples 1 and 2, we've talked about how children might use groups of 5 to help them figure out the total. Let's consider another example. In example 3, children might also use that approach. However, in this case one or two children might use 10 as a reference point in the following way. The child might reason that there are 10 beads on each rod and one on each rod has not been moved, so the number shown is 2 less than 10 plus 10, two less than 20. But what is that? The child will most likely figure out how many that is by counting back, "19, 18."

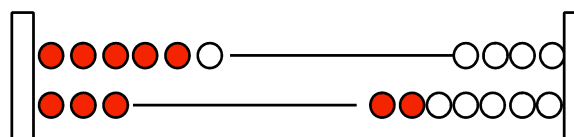
I want to say something about these solutions that involve some counting—counting on and counting back. Earlier I said that a goal is that children use the color configurations of the beads to figure out how many there are without counting, explicitly without counting everything by ones. Many of the ways children reason on these tasks involve some counting. Why is that? It's because children do not yet "just know" the number combinations for small numbers—in particular for numbers between 10 and 20. Coming to know these is a goal of the SN sequence. It is a result of using these instructional activities. Therefore, the teacher has to be careful to not insist on no counting at all. Rather, the teacher's task is to continue to encourage children to use approaches that do not involve counting all, by ones (while yet realizing that a few children will need to do so for some time yet).

Examples of tasks to not use. Here we show two examples of *unproductive* tasks. In other words these are types of tasks not to use.

Example 1:



Example 2:



Why do we call these unproductive? The reason is that there are no easy ways to capitalize on the color configurations to figure out the total. Even those children that "see" the group of 5 red beads on one rod will most likely have to count everything beyond the 5 by ones to figure out how many in all. It is important that the teacher understand this point so that she can be deliberate in using only those tasks that are productive for advancing her agenda. Careful teacher planning is essential. Teachers that have used this instructional sequence successfully typically plan the tasks they will use in a lesson by thinking about how children will reason to figure the total.

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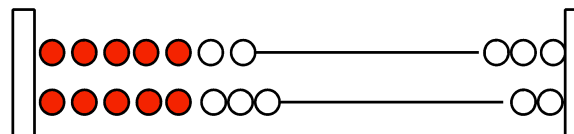
Alternative way to pose the task. In the instructional activity we have just been talking about, children figure out how many people are on the bus when the configuration is specified. The teacher specifies the configuration by showing her rack and telling students, “Make yours look like mine.” There’s another way the teacher can specify the configuration. Instead of showing her own rack she can *tell students* how many people are on the upper deck and how many are on the lower deck of the bus. Children move the beads on their racks to show that information and then figure out how many people are on the bus in all.

Example: The teacher says, “There are 7 people on the upper deck of the bus and 8 people on the lower deck. Show that on your rack and then figure out how many people are on the bus.”

When the task is posed in this way, children initially have to think about how to show the number of people on each deck of the bus. In this example, they have to think about how to show 7 on top and 8 on the bottom and only then think about the total number of people on the bus.

Now I want to be sure to clarify that this is *not* intended as an addition task. We do *not* expect that children will think, “What’s $7 + 8$?” And it’s important that the teacher not think of this as posing an addition task. At this point children do not know what $7 + 8$ is. In fact knowing sums such as this is the *result* of this entire sequence. That is the main purpose of Part II of this sequence, which we will discuss later. Instead, this task asks children to use the color configuration of the beads to figure out the totality without necessarily thinking of it as 7 and 8 more.

Posing the task in this way is simply an alternative way to ensure that all children have the same configuration on their racks; that they are all looking at the same thing, as they try to figure out how many people are on the bus. In this case, their racks will look like this.



You can see that one feature of posing the task in this way is that children have to figure out how to show numbers up to 10 because the number on each deck is necessarily 10 or less. So posing the task in this way has the potential to be particularly productive for those children that are still working on developing 5- and 10-referenced relationships for numbers to 10. Remember that the P&P instructional sequence was designed specifically for numbers to 10 so we would expect that many children will already have well-developed concepts for these small numbers. A number of children will show 7 on the upper deck by moving the 5 red beads as a group and 2 white beads as a group. Nevertheless, this task provides additional opportunities for those children that are still working on developing these 5- and 10-referenced relationships of numbers to 10 to do so.

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Once children have made the configuration on their racks and had time to figure out the total, the teacher conducts a class discussion in which children explain how they figured out the total. The ways they reason are similar to those we have already discussed for the earlier version of this task.

As a pragmatic point, I want to clarify that even though the teacher did not show her own rack when posing this task, it is useful for her to do so as the discussion progresses so that she can use gestures to help elaborate and clarify the explanations children give.

Before leaving our discussion of this activity I want to make one additional point. We have explained that an essential feature of this task is that every child's rack looks exactly the same. And we have described two different ways that the teacher can pose the task to ensure that children's racks are all alike. Most teachers we have worked with have chosen to pose the task in both of these ways, often intermingling the modes of presentation within a single lesson. In this way, they capitalize on the merits of both ways for the benefit of the students.

3a. Partner Version of Activity 3

All of the activities we have described so far are whole-class, teacher-led activities. Partner work provides a useful contrast. The activity we describe here is a partner version of the previous whole class activity. Here, children work with a partner to take turns posing tasks to each other in one of the two ways used in the previous whole class activity.

- One child uses his rack to show some configuration. The partner's task is to make his rack look exactly the same, to figure out how many are on the bus, and to explain how he figured it out.

Or

- One child tells his partner how many people are on the upper deck of the bus and how many are on the lower deck of the bus. The partner's task is to show that on his rack, to figure out how many are on the bus, and to explain how he figured it out.

The first time we tried this activity we were unsure if it would be productive. We were concerned that children might show their partner some configurations that are not useful for learning. We were pleasantly surprised. For the most part, the configurations children created were reasonable. In addition, we were amazed to see how many different tasks they posed to one another in a short amount of time. Whereas in the whole class activity, a considerable amount of time is devoted to discussing each task, in partner work only one person is explaining so much less time is involved in each task.

As a point of clarification, when we explain the partner work, we tell students that the child that poses the task should know the answer to the task he poses. He should know how many people are on the bus. As a result, children typically choose configurations and make number choices that are appropriate for their current levels of understanding. For example,

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the least sophisticated students choose tasks such as all 10 beads on the top, or all of the red beads, or 5 on one rod. The more sophisticated students, on the other hand, often choose more complex tasks that require more reasoning and more explanation, such as, 7 on the top and 9 on the bottom,

It is essential that teachers understand an important principle that undergirds our approach to partner work. It is that each child is partnered with another child at a similar level of conceptual understanding. Partners should be close to each other conceptually. Little learning takes place when students of very different conceptual understandings are partners. In our approach, partner work is not a situation for tutoring. Our more sophisticated students need to be partnered with others that will challenge them. And our least sophisticated students need to work with others that don't tell them what to do.

When will you use partner work? Teachers often like to follow whole-class activities with partner work. For example, the whole class activity we have just described previously followed by this partner activity works well. Most teachers do not conduct a discussion following partner work. Those that do, keep the discussion very brief, typically asking one or two students to describe something that happened in their group.

What might children do and learn? What might children do and learn by engaging in the activities with the arithmetic rack that we have described so far? We have already said that children will do a variety of different things and reason in a variety of ways when moving the beads and figuring out how many people are on the bus. These ways are reflective of the children's current concepts of number. Of course it is also the case that the ways children move the beads and reason change over time. These changes indicate learning. For example, a child that initially moves beads by ones and counts by ones to figure out how many people are on the bus will shift to moving groups of beads based on color configurations. In the process, such children develop 5 as a composite unit. Further, because of the support provided by the rack, children also begin to organize quantitative material in groups related to 5's, 10's, and doubles. And some will learn how the teen numbers relate to 10, though many children will not develop that relationship until further in this SN sequence. At this stage most children rely heavily on being able to use the rack and the support it provides because of the arrangement of the beads.

In summary, the learning that occurs from the instructional activities we have described thus far include that children:

- Shift from moving beads by ones to moving groups of beads based on color configurations.
- Develop a composite unit of 5.
- Organize quantitative visual material in groups related to 5's, 10's, and doubles.
- Learn how teen numbers relate to 10.

A further goal of the sequence is that children eventually become able to reason entirely mentally, that is, in the absence of the rack as a support. One step along the way is that they develop mental imagery for numbers to 20, imagery that is 5-referenced, 10-referenced,

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and doubles-referenced. The next instructional activities in the sequence are designed for that purpose.

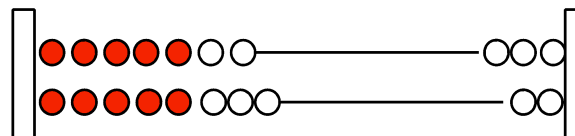
4. Rack Flash

- The teacher uses the flashing approach to display a rack configuration. The task is to try to figure out how many people are on the bus “without counting.”
- Children do not have their ARs for this task, though one or two may need to have them.
- The subsequent discussion focuses on grouping strategies.

Rack flash is designed specifically to promote mental imagery. In this activity the teacher flashes (not shows) some configuration on her AR. The children’s task is to figure out how many people are on the bus. And, as in the previous activities, they are to try to do so without counting (by ones). During this activity the children do not have their ARs so they have to rely on the images they see flashed on the screen.

Example:

The teacher flashes the following image on the screen for 2 to 3 seconds, then removes the image and leaves it off for approximately 5 seconds.



Then she flashes it again and removes the image again, and repeats the process for up to three flashes. The flashing is followed by a class discussion in which children explain how they figured out how many people are on the bus. The discussion begins while the visual material is not visible. Typically, the teacher shows the image as the discussion progresses so that children can make sense of what they hear others say.

As we have explained in earlier videos that describe flashing activities, the purpose of flashing is to promote mental imagery, in this case imagery of the rack. How does that imagery develop? It develops because during the flashing activities children have access to the visual material, in this case the rack, only briefly but are nevertheless expected to solve the task posed. That means that they have to figure out how to use the image even during the time it is not visible. Some teachers say to the students, “Take a picture with your mind.” We know that children organize the visual material mentally based on the ways of reasoning they developed when using the rack themselves in the earlier activities. We know this because of the remarks they make in the subsequent class discussion when they explain how they figured out the number of people on the bus. For example, a child might say, “First I saw all the red beads, and then the next time you flashed it I saw the 2 white beads on the top. Then the next time I saw the 3 white beads on the bottom.” The child might then go on to say, “So I knew there were 10, and then I counted, 11, 12, 13, 14, 15.”

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Remarks such as these provide evidence that children's mental images are organized around 5's, 10's and, in some cases, doubles. As with all of the activities in this sequence, the discussion focuses on strategies that use groups of beads to figure out the total, with the full understanding that some counting, such as counting on, is often necessary.

There is one additional important remark about this activity and it relates to an *underlying principle* of our instructional approach. The principle is this. *Every child should have a personally meaningful way to participate in each task.* That means that we have to accommodate the expanse of conceptual development that is present in any classroom. We have to make it possible for the less sophisticated student to have a way to participate while at the same time make it possible for the more sophisticated student to advance in his learning. We have described this task in terms of how it provides an opportunity for students to advance from using physical materials to using mental material as the basis for reasoning. However in almost every class there will be one or two students that still require the physical material to participate meaningfully. Therefore we recommend that the teacher ensure that those students that need to have a rack have one to use. What will happen is that as the teacher repeatedly flashes the image on the screen, the children that have a rack to use will make their rack look like the image flashed and then use it to reason with. If more than 2 or 3 students in the class require a rack, use of this activity is premature.

4a. Partner Version of Rack Flash

In a partner version of rack flash, each pair of students has a rack. They take turns posing tasks to each other as follows. One child creates a configuration on the rack and shows it to the partner. The partner's task is to figure out how many people are on the bus and explain how he figured it out.

The difference between this partner activity and the one we described earlier is that here only the child that creates the task has a rack. The partner's task is to figure out how many people are on the bus without having his own rack to reason with. Of course, those children that still need a rack to use often touch the beads as they figure it out.

We have found that it is not necessary that the child posing the task mimics rack flash to the extent that he actually shows and hides the rack repeatedly. It is sufficient that he shows the rack to the partner.

Teachers often use this partner activity in the same lesson as the whole-class rack flash activity. Many teachers find that following a whole class activity with partner work that is similar in nature provides a useful change of pace for the children while continuing to promote the intended learning goals of the lesson.

5. Rack Bingo

- The teacher flashes a configuration on her rack. Children cover the number shown on their bingo card.

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- Discussion is kept to a minimum to keep the game moving at a reasonable pace.

Another activity that provides some variation in format to rack flash is one that we call rack bingo. In this activity each child has a bingo card and a supply of chips. To play the game, the teacher flashes a configuration on her rack. Children cover the number the teacher flashed on their bingo card. The game continues in this fashion until one or more children achieve bingo.

One feature of this game is that many tasks are posed in a short amount of time, else no one would ever achieve bingo. For this reason discussion is kept to a minimum. Most teachers call on only one child to tell how many they saw and to give a very brief explanation of how they figured it out. You can see that this activity is similar to rack flash. The primary difference is that there is very little discussion here. Only one person is called on to explain how they figured out each task, whereas in rack flash the teacher will typically call on a variety of students and will involve all students in the discussion by asking children what they understand about an explanation. In other words, the purpose of rack flash is to work hard at advancing conceptual development. Rack bingo is appropriate as an activity only after most students are already adept at figuring out the total flashed.

There are some pragmatic considerations when playing this game. First, the teacher has to remember that in a bingo game, the game cards the children have should not all be the same. Some teachers create a variety of bingo cards themselves that they hand out to students. Other teachers we have worked with have children initially create their own bingo cards. One easy way to do so is to devote a part of one lesson to this purpose. Each child has a *blank* bingo card. (For example, it might be a 3 by 3 grid.) The teacher poses a series of *rack flash* tasks. Each task is discussed as we have described previously. In addition, each child records the number shown on each rack flash task anywhere they like on their bingo card. In this way the bingo cards are individualized and each child has his or her own bingo card now ready for use with the bingo activity.

A second pragmatic consideration is this. It is not uncommon for first grade students to accidentally “mess up” their bingo card during play. They might just accidentally brush some of the chips off of their card. Teachers we have worked with have addressed this issue by using the white board to keep a running list of the numbers flashed. Children can refer back to this list, if they need to reassemble the chips on their card.

A third pragmatic consideration refers to dealing with the emotions of “not winning.” A number of teachers we have worked with deal with this issue by simply continuing the game until everyone has achieved bingo.

6. Rack Flash Variation

In this variation children are asked to tell three things:

1. How many people are on the upper deck.
2. How many people are on the lower deck.

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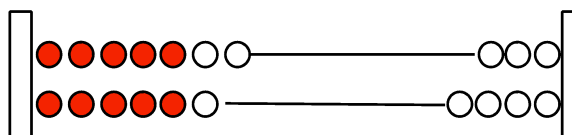
3. The total number of people on the bus.

Once most children are adept at figuring out the total shown in the rack flash task, the teacher can choose to use the variation we describe here. In this variation children are asked to tell how many they saw on the upper deck of the bus, how many they saw on the lower deck of the bus, and the total number of people on the bus. In other words, in this task children are asked to focus on more details. Remember that this is a flashing task. Children have to use their mental images to think about the two individual quantities and also about the totality shown. This variation on rack flash can be thought of as a natural extension of the earlier rack flash task, one where children are asked to pay attention to more details. A typical discussion follows this task. Children explain what they figured out and how they figured it out. The teacher gets the class involved in the discussion by asking what they understand about someone's explanation, calling on students that have different explanations, and so on.

Drawing attention to the number of people on the upper deck and the lower deck of the bus in addition to the total creates an opportunity for the teacher to initiate a discussion about the sum of two single-digit quantities, *if* she chooses to do so. In this way, this task previews Part II of the instructional sequence. In addition, by using 10 on one deck of the bus with a single-digit quantity on the other deck, children begin to develop the relationship between a single-digit number and the corresponding teen number.

Example:

In this example the teacher flashes the configuration shown here, using the typical approach of flashing the image up to three times.



The children's task is to:

- Say how many people are on the upper deck of the bus.
- Say how many people are on the lower deck of the bus.
- Say how many people are on the bus in total.

And to explain how they figured it out.

The image is left off as the discussion begins.

To continue this example, suppose a child, Antonio says, "I saw 13 people on the bus. I saw all of the red beads. That's 10, and then I saw 2 white beads on top and 1 white bead on the bottom. That's 3 more. 10 and 3 more is 13." Here Antonio explained how he figured out the total but he did not say how many were on the top and how many on the bottom. The teacher might then say, "How many did you see on the top and how many on the bottom?"

- What do you understand about what Antonio is saying?
- Who figured it out like Antonio?
- Who figured it out a different way?

This verbal exchange with Antonio gives the teacher the opportunity to initiate a discussion *as if* this task was about the sum of 7 and 6, if she so chooses. She might say, “Antonio said he saw 7 on the upper deck. He saw the 7 as 5 and 2 more. He saw 6 on the lower deck. He saw the 6 as 5 and 1. The two 5’s make 10. The 2 white beads on the top and the 1 white bead on the bottom make 3. Antonio said that he knows that 10 and 3 make 13.”

This is a small group version of the variation on rack flash. Children take turns posing tasks to each other. One child creates a configuration on the AR and shows it to the partner. The partner's task is to tell three things:

- And also to explain how he figured it out.

As with the partner task discussed earlier, many teachers do not have a discussion following partner work. Those that do typically call on one or two children to describe something that happened during partner work.

Here we show the general timeline for the entire instructional sequence.

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PART II Becoming efficient with adding and subtracting using grouping strategies prompted by the rack

PART III Becoming efficient with additive tasks without the rack

In these notes we have provided an overview of the Structuring Numbers instructional sequence and described in detail the instructional activities that comprise Part I of the sequence, becoming efficient with reading the rack. Here we show a general timeline for the entire sequence and a detailed timeline for Part I. As the timeline shows we recommend devoting approximately three and a half weeks to part I of the sequence and approximately five and a half weeks to the remaining parts of the sequence.

Now let's take a look at the detailed timeline for part I.

Detailed Timeline for Part I of the Structuring Numbers Instructional Sequence

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9
PART I Becoming efficient with reading the rack								
Introductory activities								
Show this many people on the bus								
How many people – make configuration specified								
Rack flash								
Rack bingo								
Rack flash variation								

The first two activities of Part I are initiating activities and hence are used only once, as we have described previously. As the timeline shows, the other activities are used repeatedly over several weeks. The overlap in the timeline indicates that we intend that some of these activities be used together within the same lesson. For example, the third activity in which students make the configuration on their racks that the teacher specifies and then figure out the total begins within the first few days and continues through the second week and possibly into the third week. At the same time, the activity called rack flash begins within the second week and continues through the third week.

To clarify what we mean, the following is a possible lesson outline the teacher might use in the second week or early in the third week:

Lesson Outline:

Activity 1: How many people on the bus – children make the configuration the teacher specifies and figure out the total

Activity 2: Partner version of activity 1.

Activity 3: Rack flash

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The rationale for including several different instructional activities that have differing conceptual demands within the same lesson is that by doing so the teacher ensures that the more conceptually sophisticated students within the class continue to have learning opportunities at their level. At the same time other students within the class continue to progress in their own conceptual development because students always participate in ways that are personally meaningful to them. We want to be clear that we are not misunderstood on this point. Every student participates in every activity within a lesson. But each student should have a personally meaningful way to do so. In some cases this may mean that some of the less conceptually advanced students will have a rack to use during an activity such as rack flash while the majority of the students do not have a rack to use.

There are **two fundamental principles** that undergird our approach to instruction.

1. **Every child should have a personally meaningful way to participate in every activity.**
2. **Every child should have the opportunity to advance in his learning in every lesson.**

Planning for several different activities within the same lesson makes it possible for both of these principles to be met at the same time.

As you can see from this description and this lesson example there is a gradual progression through the instructional activities that make up Part I of the sequence. New activities are added to the lessons while prior activities continue and gradually drop out when the majority of students are no longer learning anything new from them.

Finally we reiterate what we have said repeatedly when giving the detailed discussion of the activities. The activities in this sequence are designed to promote conceptual development, such as five as a composite unit, five-referenced, ten-referenced, and doubles-referenced concepts of numbers to 20, and the relationship between 10 and the teen numbers. The goal is *not* that students learn “to do” the activities. Teachers have to make judicious decisions about how and when to move forward and when to step back and devote more time to something. The most important thing the teacher has to attend to is how her students are participating and what that indicates about how they are reasoning.

Next Videos and Notes

This is the first of four sets of notes, one set to accompany each of four video groups. The next three sets of notes accompany the next three video groups:

- Structuring Numbers II
- Structuring Numbers III
- Teacher’s role in implementing the SN sequence

Structuring Numbers II and Structuring Numbers III each describe another part of the instructional sequence. The last video group addresses the teacher’s role in successfully

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implementing the Structuring Numbers sequence. It is devoted to two main issues. One issue relates to planning lessons, including selecting which activities to combine to comprise a lesson and, for any specific activity, selecting which tasks to pose to enhance the learning opportunities for students. The second issue is how to orchestrate productive class discussions. Examples are used to illustrate these issues. We note that this fourth video group can be viewed meaningfully at any time. It is not necessary to first view all three of the video groups that describe the details of the instructional activities.